

Yue Yu

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RESEARCH INTEREST

Integrated photonic circuits, optomechanics, MEMS, nonlinear photonics, surface acoustic waves, bound states in the continuum

EDUCATION

- | | |
|-----------------------|--|
| Jan. 2025 – Present | Postdoc, Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, Berkeley, California, United States |
| Sep. 2023 – Dec. 2024 | Postdoc, Viterbi School of Engineering, University of Southern California, Los Angeles, California, United States |
| Aug. 2017 – July 2023 | Ph.D., Department of Electronic Engineering, The Chinese University of Hong Kong, Hong Kong SAR |
| Sep. 2013 – June 2017 | B.S., Department of Optical and Electronic Information, Huazhong University of Science and Technology, Hubei, China |

PROFESSIONAL ACTIVITIES

Journal reviewer

- Optics Express
- Journal of Lightwave Technology
- Journal of the Optical Society of America B
- Communications Engineering

PUBLICATIONS

Journal Papers († denotes co-first authors)

1. **Yue Yu**[†], Ran Yin[†], Ian Anderson, Yinan Wang, Jack Kramer, Chun-Ho Lee, Xinyi Ren, Michelle Povinelli, Dan Wasserman, Ruochen Lu, and Mengjie Yu, “Room-temperature low-noise optomechanical platform on thin film lithium niobate”. (In preparation)
2. Ran Yin[†], **Yue Yu**[†], Chunho Lee, Ian Anderson, Xinyi Ren, and Mengjie Yu, “Fundamental frequency noise of thin film lithium niobate platforms”. (In preparation).
3. Silvia Guadagnini, Zarko Sakotic, **Yue Yu**, Mengjie Yu, Daniel Wasserman, Michelle L. Povinelli, “Suspended, polarization-dependent, subwavelength-perforated metal absorber for mid-infrared bolometry”, *Nano Letter*, Oct. 2025 (accepted).
4. Shaoyuan Ou, Kaiwen Xue, Lian Zhou, Chun-ho Lee, Alexander Sludds, Ryan Hamerly, Ke Zhang, Hanke Feng, **Yue Yu**, Reshma Kopparapu, Eric Zhong, Cheng Wang, Dirk Englund, Mengjie Yu, and Zaijun Chen, “Hypermultiplexed Integrated-Photonics-based Tensor Optical Processor”, *Science Advances* 11, eadu0228, June 2025. (accepted)
5. Xinyi Ren, Chun-Ho Lee, Kaiwen Xue, Shaoyuan Ou, **Yue Yu**, Zaijun Chen, and Mengjie Yu, “Photorefractive and pyroelectric photonic memory and long-term stability in thin-film lithium niobate microresonators”, *npj nanophotonics* 2, 1, Jan. 2025.
6. **Yue Yu** and Xiankai Sun, "Surface acoustic microwave photonic filters on etchless lithium niobate integrated platform," *Laser & Photonics Reviews* 2300385, 2024.
7. **Yue Yu** and Xiankai Sun, "Etchless photonic integrated circuits enabled by bound states in the continuum: tutorial," *Journal of the Optical Society of America B* 40 (11): 2801–2808, Nov. 2023.
8. Zejie Yu, He Gao, Yi Wang, **Yue Yu**, Hon Ki Tsang, Xiankai Sun, and Daoxin Dai, "Fundamentals and applications of photonic waveguides with bound states in the continuum," *Journal of Semiconductors* 44 (10): 101301, Oct. 2023.
9. Xiao-Jing Liu[†], **Yue Yu**[†], Di Liu, Qi-Long Cui, Xiao-Zhuo Qi, Yang Chen, Guang-Yuan Qu, Li Song, Guo-Ping Guo, Guang-Can Guo, Xiankai Sun, and Xi-Feng Ren, "Coupling of photon emitters in monolayer WS₂

- with a photonic waveguide based on bound states in the continuum," [*Nano Letters* 23 \(8\): 3209–3216, Apr. 2023.](#)
10. Yuan Li, Zunyue Zhang, Yi Wang, **Yue Yu**, Xuotong Zhou, Hon Ki Tsang, and Xiankai Sun, "Inverse-designed linear coherent photonic networks for high-resolution spectral reconstruction," [*ACS Photonics*, Jan. 2023. DOI: 10.1021 \[featured as cover article\]](#)
 11. **Yue Yu**, Xiang Xi, and Xiankai Sun, "Observation of mechanical bound states in the continuum in an optomechanical microresonator," [*Light: Science & Applications* 11, 328, Nov. 2022.](#)
 12. **Yue Yu**[†], Zejie Yu[†], Zunyue Zhang[†], Hon Ki Tsang, and Xiankai Sun, "Wavelength-division multiplexing on etchless lithium niobate integrated platform," [*ACS Photonics* 9 \(10\): 3253–3259, Oct. 2022.](#)
 13. Huade Mao[†], **Yue Yu**[†], Yu-Xuan Ren, Ka Yan Chan, Jiqiang Kang, Xiankai Sun, Edmund Y. Lam, and Kenneth K. Y. Wong, "Neural optimizer for inverse design of complex-modulated hologram implemented by plasmonic metasurfaces," [*Advanced Photonics Research* 4 \(1\): 2200085, Jan. 2023.](#)
 14. Fan Ye, **Yue Yu**, Xiang Xi, and Xiankai Sun, "Second-harmonic generation in etchless lithium niobate nanophotonic waveguides with bound states in the continuum," [*Laser & Photonics Reviews* 16: 2100429, Mar. 2022.](#)
 15. **Yue Yu**, Lai Wang, and Xiankai Sun, "Demonstration of on-chip gigahertz acousto-optic modulation at near-visible wavelengths," [*Nanophotonics* 10 \(17\): 4323–4329, Dec. 2021.](#)
 16. **Yue Yu**, Zejie Yu, Lai Wang, and Xiankai Sun, "Ultralow-loss etchless lithium niobate integrated photonics at near-visible wavelengths," [*Advanced Optical Materials* 9 \(19\): 2100060, Oct. 2021.](#)
 17. Huade Mao[†], Yu-Xuan Ren[†], **Yue Yu**[†], Zejie Yu, Xiankai Sun, Shuang Zhang, and Kenneth K. Y. Wong, "Broadband meta-converters for multiple Laguerre-Gaussian modes," [*Photonics Research* 9 \(9\):1689–1698, Sep. 2021.](#)
 18. **Yue Yu**, Zejie Yu, and Xiankai Sun, "Nonmetallic broadband visible-light absorbers with polarization and incident angle insensitivity," [*IEEE Photonics Journal* 12 \(6\): 2200807, Dec. 2020.](#)

Conference Paper

1. Silvia Guadagnini, Zarko Sakotic, **Yue Yu**, Mengjie Yu, Dan Wasserman and Michelle L. Povinelli, "Suspended Subwavelength-Perforated Metal Absorber for Mid-Infrared Bolometry", CLEO 2025.
2. Xinyi Ren, Chun-Ho Lee, Reshma Kopparapu, Clayton Cheung, Lian Zhou, **Yue Yu**, Ran Yin, Zaijun Chen, and Mengjie Yu, "Few-cycle Pulse Generation on the Thin-film Lithium Niobate Platform", CLEO 2025.
3. Ran Yin[†], **Yue Yu**[†], Xinyi Ren, Chun-Ho Lee, Yuanhao Liang, Ian Anderson, Jack Kramer, Ruochen Lu, Mengjie Yu, "Intrinsic Frequency Noise of the Thin Film Lithium Niobate Platforms", CLEO 2025.
4. **Yue Yu**[†], Ran Yin[†], Ian Anderson, Jack Kramer, Chun-Ho Lee, Xinyi Ren, Clayton Cheung, Ruochen Lu, and Mengjie Yu, "Room-temperature optomechanical-resonator-based thermal sensor on thin-film lithium niobate", CLEO 2025.
5. Xinzhou Su, Chun-Ho Lee, Xinyi Ren, Zile Jiang, Huibin Zhou, Yue Zuo, Shaoyuan Ou, Reshma Kopparapu, **Yue Yu**, Adam T. Heiniger, Moshe Tur, Zaijun Chen, Mengjie Yu, and Alan E. Willner, "Demonstration of Mid-Wavelength Infrared IM/DD Communications Using Air-suspended Thin-Film Lithium Niobate Intensity Modulator", OFC 2025.
6. Clayton Cheung, Xinyi Ren, Chun-Ho Lee, Reshma Kopparapu, **Yue Yu**, Zaijun Chen, and Mengjie Yu, "Lithium Niobate Chip-Based Ultrafast Optical Signal Processor", CLEO 2024, SM4L.5.
7. Xinyi Ren, Chun-Ho Lee, Kaiwen Xue, Shaoyuan Ou, **Yue Yu**, Zaijun Chen, and Mengjie Yu, "Long-lived photorefractive and pyroelectric effects in thin film lithium niobate microresonators", CLEO 2024, STu3E.1.
8. **Yue Yu** and Xiankai Sun, "Surface acoustic microwave photonic filters on etchless lithium niobate integrated platform," CLEO 2023, San Jose, CA, USA, May 2023.
9. Yuan Li, Zunyue Zhang, Yi Wang, **Yue Yu**, Xuotong Zhou, Hon Ki Tsang, and Xiankai Sun, "Inverse-designed linear coherent photonic networks for high-resolution spectral reconstruction," CLEO 2023, San Jose, CA, USA, May 2023.
10. **Yue Yu**, Xiang Xi, and Xiankai Sun, "Observation of mechanical bound states in the continuum in an optomechanical microresonator," **Frontiers in Optics 2022**, Rochester, NY, USA, Oct. 2022. [postdeadline]

11. **Yue Yu**[†], Zejie Yu[†], Zunyue Zhang[†], Hon Ki Tsang, and Xiankai Sun, "Wavelength-division multiplexing on etchless lithium niobate integrated platform," **Frontiers in Optics 2022**, Rochester, NY, USA, Oct. 2022.
12. **Yue Yu**, Zejie Yu, Lai Wang, and Xiankai Sun, "Ultralow-loss etchless lithium niobate integrated photonics at near-visible wavelengths," **CLEO 2022**, San Jose, CA, USA, May 2022.
13. Fan Ye, **Yue Yu**, Xiang Xi, and Xiankai Sun, "Second-harmonic generation in etchless lithium niobate nanophotonic waveguides with bound states in the continuum," **CLEO 2022**, San Jose, CA, USA, May 2022.
14. **Yue Yu**, Zejie Yu, and Xiankai Sun, "Etchless lithium niobate integrated photonics," **International Symposium on Lithium Niobate Optoelectronics 2021**, Shanghai, China, Oct. 2021. [invited]
15. **Yue Yu**, Zejie Yu, and Xiankai Sun, "Nonmetallic broadband visible-light absorbers with polarization and incident angle insensitivity," **CLEO 2021**, San Jose, CA, USA, May 2021.
16. **Yue Yu**, Lai Wang, and Xiankai Sun, "Demonstration of on-chip gigahertz acousto-optic modulation at near-visible wavelengths," **CLEO 2021**, San Jose, CA, USA, May 2021.